

**ADD ONS OVERVIEW**

Deos Heartbeat Add Ons

UKDEOS

Location, Daily & Weekly Insight, Owned Data, Tickets, Report Engagement, and Trust Access

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Relates to	Deos Heartbeat Lite, Scope of Work v2.0

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1. Overview

Deos Heartbeat is the operating system for UKDEOS: one central platform where you run reporting, fleet health, and everything that follows. Deos Heartbeat Lite provides the foundation and its first two modules, Reporting Management and Fleet Health. The add ons in this document are further modules that plug into that same foundation, adding location and location history, daily and weekly history, permanently owned data, open tickets and ticket history, report engagement insight, and direct access for your trusts.

Each add on sits inside the same platform, uses the same login, and draws on data your vans already send to Grafana. There is no new hardware and nothing extra to install on the vans.

At a glance

Live Satellite Fleet Map: see every van on a satellite map of the UK, colour coded by status.

Location History: go back to any day and see where a van was, where it moved, and how it performed at every site.

Daily & Weekly Dashboards: daily and weekly performance for the whole fleet, each trust with its vans, and each individual van.

UKDEOS Owned Data: your fleet's full performance history, owned by UKDEOS and kept permanently, with month on month and year on year views.

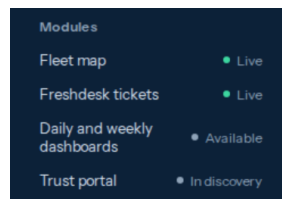
Freshdesk Tickets: every open ticket across the fleet, plus ticket history alongside each trust's and van's performance.

Email / Report Engagement Dashboard: see who is opening your reports, which trusts are engaged, and which recipients have gone quiet.

Trust Portal: secure access for each trust to its own vans, dashboard, and fleet map (subject to discovery).

Add ons confirmed by the end of discovery (Week 2) are built alongside the core platform, and most can be delivered within the same 10 week programme. Add ons requested later are scheduled as a separate delivery after go live. The Trust Portal is the exception: because it gives external NHS organisations access to UKDEOS systems, it requires a discovery phase with your trusts before it can be scoped.

About the images in this document: all screenshots are demo images from a design mockup, showing sample data. They illustrate the intended layout and features. The final design will be shaped with you in the UI design sessions, and the live platform will show your real fleet data.



Demo image: In Deos Heartbeat, add ons appear as modules alongside the core platform.

2. Live Satellite Fleet Map

Every van in the UKDEOS network already reports its GPS location through Grafana via its onboard modem. The fleet map turns that data into a live picture of your entire fleet on real satellite imagery, presented as an interactive globe in the style of Google Earth.



Demo image: The live fleet map, with every van colour coded by status. Satellite imagery replaces this base map in the build.

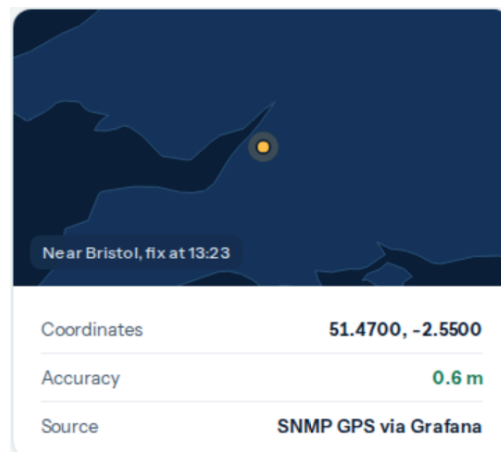
2.1 What You Get

- Globe style satellite view: the map opens on a 3D globe centred on the UK, with real satellite imagery rather than a flat coloured map
- Smooth zoom from the globe down to a national view, a region, and individual screening sites, with high resolution satellite imagery at every level
- Switch between satellite and street map views at any time
- Colour coded markers matching fleet health status: green (online), amber (degraded), red (offline)



- Nearby vans grouped at national zoom and separated as you zoom in, so busy areas like London stay readable
- Click any van to open its full van detail view
- Filter the map by trust or by status, for example show only offline vans
- GPS health panel: flags any van reporting no location fix or poor location accuracy, which is often an early sign of a modem fault
- Accuracy indicator: hovering over or clicking any van shows its reported GPS accuracy in metres and when its location was last updated. Positions with lower accuracy are also shown with a shaded circle representing the likely area, so an approximate location is never mistaken for an exact one

Location accuracy: each van's position comes from its modem's GPS, which reports an accuracy value alongside every position. Most vans report accuracy within a metre, but accuracy can deviate depending on the van's surroundings, for example near tall buildings, in covered or multi storey car parks, or when a modem is struggling. Recent Grafana data illustrates this: most vans reported accuracy under one metre, one reported around 10 metres, one around 500 metres, and one had no GPS fix. The map shows lower accuracy positions with a shaded accuracy circle rather than an exact pin, flags them in the GPS health panel, and does not plot vans with no fix.



Demo image: Each van's location card shows its coordinates, GPS accuracy, and data source.

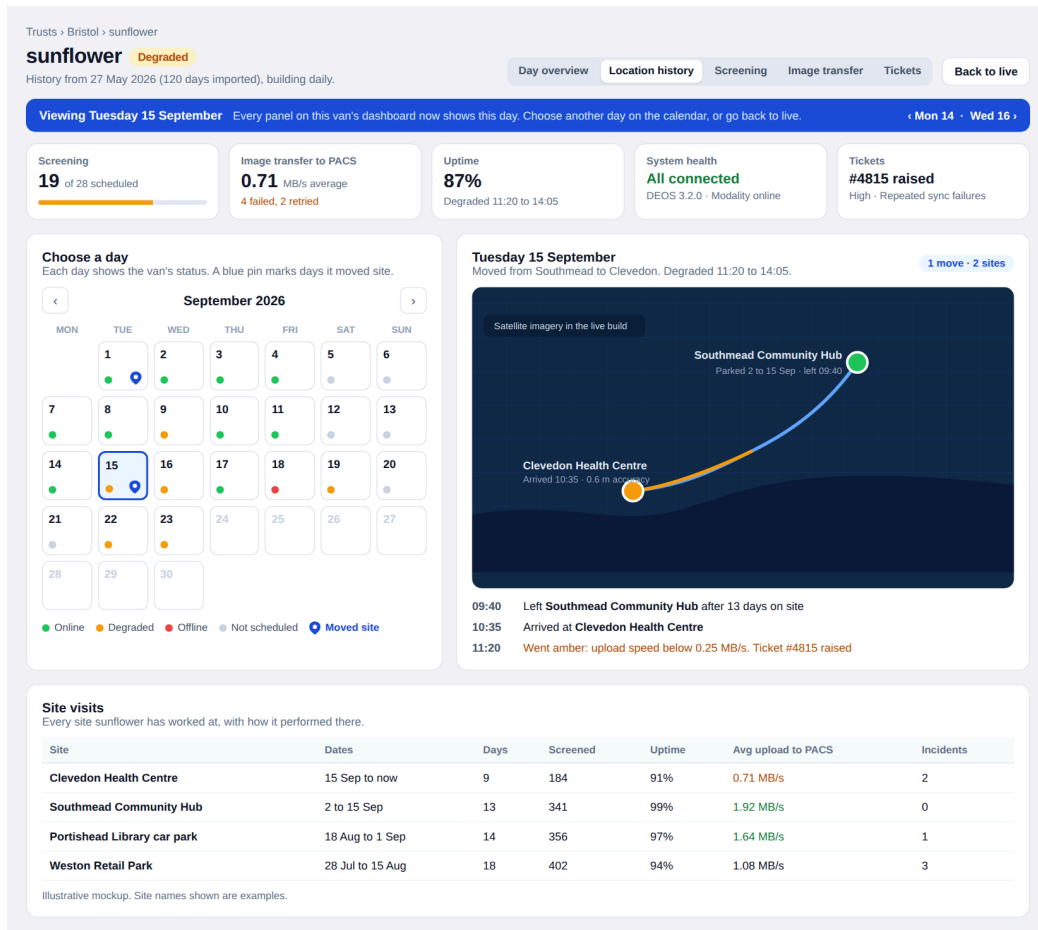
2.2 What It Helps You Do

Situation	With the Fleet Map
A trust calls about a van problem	See exactly where the van is and its live status before you pick up the phone
A van goes offline	See where it is parked and whether other vans nearby are affected
A van loses its GPS signal	It is flagged in the GPS health panel, prompting a modem check before it becomes a bigger fault
Showing NHS England or a buyer the fleet	A live national map of your screening infrastructure, colour coded by health



3. Location History

An upgrade to the Live Satellite Fleet Map that lets you go back in time for any van. Open a van, choose a day on the calendar, and the whole van dashboard shows that day: where the van was and where it moved, alongside how it performed.



Demo image: A van dashboard viewing a past day, with the Location History tab open.

One important distinction

The calendar does not just show location. Choosing a day rewinds the entire van dashboard to that day. The Location History tab shows where the van was and the route it took, while every other panel shows the same day: patients screened against scheduled, image transfer to PACS, uptime, system health, and any tickets raised. Everything that happened to that van on that day is in one place.

3.1 How It Works

- The calendar sits within each van's dashboard. Every day is coloured by the van's status that day, and a pin marks the days it moved site



- Select any day and a banner confirms which day you are viewing. Step to the previous or next day, or return to live at any time
- The Location History tab shows where the van was on the map, the route it took if it moved, and a timeline of the day: when it left a site, when it arrived at the next, and when it went degraded or offline
- The van's other panels show the same day: screening, image transfer, uptime, system health, and tickets
- Site visits list every site the van has worked at, with the dates, days on site, patients screened, uptime, average upload time to PACS, and incidents at each

3.2 Why It Helps

- See exactly where a van was parked when it went offline, and what else was happening at the time, to diagnose problems quickly
- Compare how the same van performs at different sites, and spot sites with consistently poor signal
- Answer questions from trusts about a specific day, such as where the van was, how many patients were screened, and what went wrong, without searching through Grafana

3.3 History

- On launch, up to 120 days of location history is imported from Grafana, so around four months are available from day one
- From then on, location history is stored permanently in UKDEOS's own database, building every day
- GPS readings are condensed into site stays and journeys, which keeps the view fast and filters out small jumps from lower accuracy readings

Location History requires the Live Satellite Fleet Map. The day view of screening, image transfer, uptime, and health uses the daily history from the Daily & Weekly Dashboards, and tickets appear when Freshdesk Tickets is selected.

4. Daily & Weekly Dashboards

The core platform shows what is happening right now. Daily & Weekly Dashboards add history at three levels: the whole fleet, each trust with its vans, and each individual van. A simple Daily / Weekly toggle switches every view between a single day and a full week.

- Daily view: any single day, compared with the same day the previous week, since screening follows a weekly pattern
- Weekly view: the full week, compared with the previous week



4.1 Fleet Overview

- Total patients screened, screened against scheduled, and fleet uptime for the selected day or week
- Trust league table ranked by screening volume, uptime, and sync reliability

4.2 Trust Dashboards

Each trust dashboard shows combined figures across all of that trust's vans, with every van listed underneath.

- Combined worklist: total patients scheduled across the trust's vans against total patients screened, with a completion rate
- Average image upload time to PACS across the trust, and average sync speed
- Trust uptime, total studies completed, total sync failures, and total time vans were offline
- Daily chart of patients screened against scheduled for the trust
- Every van listed underneath with its own figures and ranking, each linking through to its van dashboard
- Daily or weekly trust PDF, ready to send to the screening office

Vans online

7 of 8, 1 paused

Screened today

126 of 234 scheduled

Upload to PACS

2.85 MB/s average, 0 failed

Open tickets

0 last closed Fri 18 Sep

Vans

Switch a van off to leave it out of that report. Paused vans are skipped entirely.

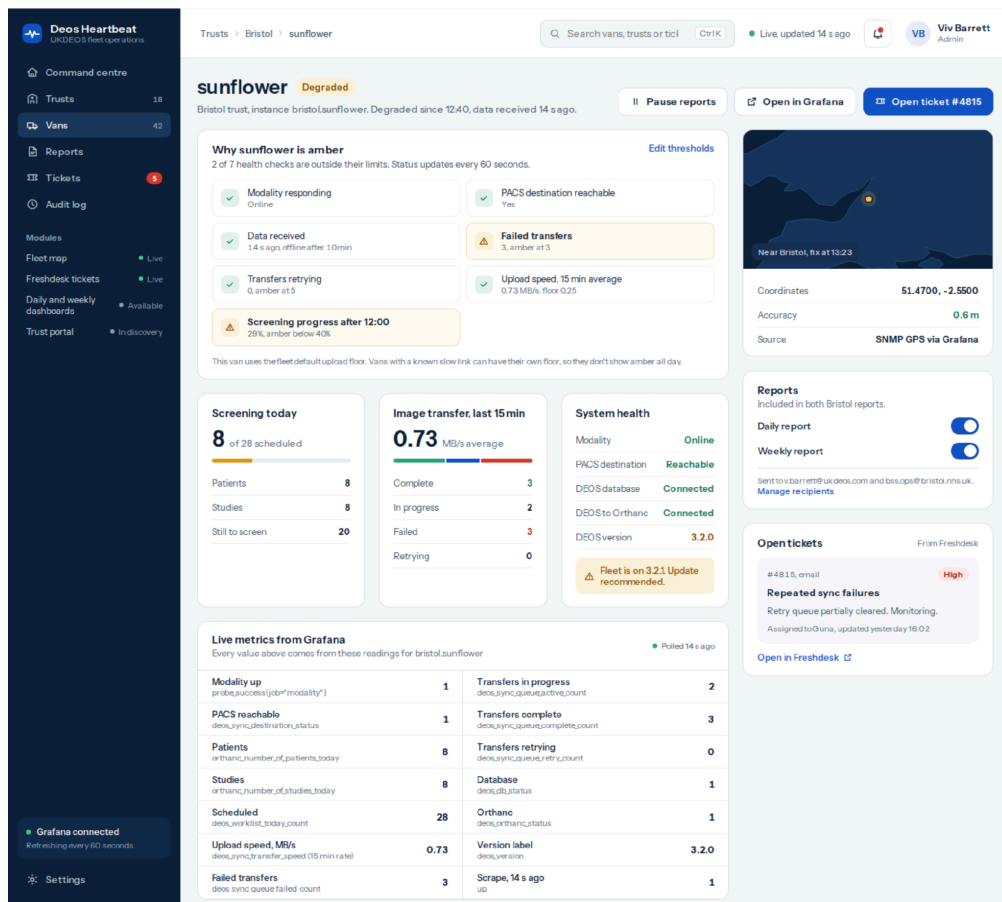
Van	Status	Screened today	Upload	Failed	Daily	Weekly	Actions
van1-daisy london.van1-daisy	Online	14 of 30 47%	3.53 MB/s	0			Pause
van2 london.van2	Online	20 of 36 56%	3.55 MB/s	0			Pause
van3 london.van3	Paused	None scheduled 0%	No data	0			Resume
van4 london.van4	Online	20 of 36 56%	1.97 MB/s	0			Pause
van5 london.van5	Online	13 of 28 46%	4.64 MB/s	0			Pause
van6 london.van6	Online	16 of 30 53%	0.91 MB/s	0			Pause
van7-stmarys london.van7-stmarys	Online	21 of 36 58%	2.29 MB/s	0			Pause
van8-violet london.van8-violet	Online	22 of 38 58%	3.05 MB/s	0			Pause

Demo image: A trust dashboard, with combined figures across the trust and every van listed underneath. Daily & Weekly Dashboards add day and week history to this layout.

4.3 Van Dashboards

Each van dashboard shows that van's own figures only.

- Patients scheduled against screened, and studies completed
- Average image upload time to PACS and sync speed trend
- Uptime, sync failures, and time offline
- A calendar on every van dashboard: choose any day and every panel shows that day's figures, with the previous or next day one click away
- If Location History is also selected, the van's location and movements for the chosen day appear in their own tab



Demo image: A van dashboard, showing the van's own figures only. With Daily & Weekly Dashboards, choosing a day on the calendar shows every panel as it was on that day.

4.4 Trust and Van Figures at a Glance

Figure	Trust Dashboard (combined across vans)	Van Dashboard (individual van)
Worklist	Total scheduled across all vans	Scheduled for this van
Patients screened	Total screened, with trust completion rate	Screened by this van, with completion rate
Studies completed	Total across all vans	This van's studies
Image upload time to PACS	Average across all vans	This van's average
Sync speed	Average across all vans	This van's average and trend
Sync failures	Total across all vans	This van's failures
Uptime	Trust average, with each van listed	This van's uptime
Time offline	Total across all vans	This van's time offline



Daily & Weekly Dashboards read from Deos Heartbeat's own nightly record of every van's figures. On launch, up to 120 days of existing history is imported from Grafana, so around four months are visible from day one, and the dashboards cover the most recent 120 days. Keeping history permanently, with month on month and year on year views, is available with UKDEOS Owned Data (Section 5).

5. UKDEOS Owned Data

Today, all of your fleet's performance history lives in Grafana. Grafana is a strong monitoring tool, but it was built for engineers watching systems in real time, not for keeping a permanent record of how your service has performed. UKDEOS Owned Data moves that history into a database owned by UKDEOS within Deos Heartbeat, where it is kept permanently and turned into clear, month on month and year on year insight.

5.1 The Limitations of Relying on Grafana

Limitation	What It Means for UKDEOS
History is deleted after 120 days	Anything older than about four months is permanently lost. There is no way to look back at last quarter's performance in full, let alone last year's
Retention can change	A configuration change or upgrade can shorten the history Grafana keeps, without notice
Built for engineers	Data is stored as raw readings every 30 seconds. Answering a simple business question, such as how a trust performed last month, needs technical knowledge
Not a reporting record	There are no month on month or year on year views, and querying long periods is slow and heavy
Not a business asset	The history sits inside a monitoring tool rather than a structured record UKDEOS can export, share, or present

5.2 How It Works

- On launch, up to 120 days of existing history is imported from Grafana, so around four months of performance is available from day one
- Every night, each van's day is summarised and stored permanently in UKDEOS's own database within Deos Heartbeat
- There is no time limit: history keeps building for as long as Deos Heartbeat runs, month after month and year after year
- The data belongs to UKDEOS. Your full history can be exported at any time
- Once stored, history is unaffected by any later change to Grafana

5.3 What You Can See

- Month on month and year on year views for the whole fleet, each trust, and each van



- Simple ranges: last 3 months, 6 months, 12 months, or all time. No technical queries
- Plain English headlines, such as "Screening up 8% on last quarter" or "Uptime 98.4%, best quarter this year"
- A year at a glance availability calendar for every van, with each day shown green, amber, or red, so recurring patterns stand out instantly
- Trust comparison over time, showing which trusts are improving and which need attention
- Monthly, quarterly, and annual PDF reports, ready for trust reviews, contract renewals, and NHS England

5.4 Grafana Compared with UKDEOS Owned Data

	Grafana Only	UKDEOS Owned Data
How far back	120 days, then deleted	Unlimited, from launch onwards, plus up to 120 days imported
Who controls it	Monitoring tool settings	UKDEOS
Format	Raw readings every 30 seconds	Daily summaries designed for reporting
Month on month and year on year	Not available	Built in
Ease of use	Requires technical knowledge	Plain English views and headlines
Export and share	Technical and partial	Full history, exportable at any time
Affected by Grafana changes	Yes	No, once history is stored

5.5 Why It Matters to NHS England and Your Trusts

- Proven reliability over time: a documented history of uptime and image delivery for every van and trust, rather than a four month snapshot
- Evidence for service levels and renewals: month by month performance that can be shown when contracts are reviewed or renewed
- Supporting national digital screening: NHS England's move to a digital platform for breast screening depends on reliable connectivity in mobile units. Year on year evidence of UKDEOS's network performance directly supports that case
- Capacity and coverage planning: long term screening volumes by trust and van show where demand is growing and where capacity is under used
- Assurance and audit: a permanent, consistent record that can be referenced in reviews, tenders, and assurance processes
- Value to the business: an owned, structured performance history is a genuine asset in any investment or acquisition conversation



UKDEOS Owned Data holds aggregate operational figures only, such as screening counts, uptime, and transfer performance. It contains no patient identifiable information. Requires the Daily & Weekly Dashboards add on.

6. Freshdesk Tickets: Open Tickets & History

Brings your Freshdesk support tickets into Deos Heartbeat, so you can see every open ticket across the fleet and every ticket's history alongside the performance and downtime it relates to. Requires the Daily & Weekly Dashboards add on.

6.1 Where Tickets Appear

Location	What You See
Tickets page	A new page listing every open ticket across the fleet, with open, pending, and urgent counts. Filter by trust, and by open, resolved, or all tickets
Heatmap tiles	A ticket count on any van tile with open tickets, so you can see at a glance that an offline van already has a ticket raised
Van detail panel	That van's open tickets, each showing subject, priority, status, source, and when it was last updated
Van dashboards	The van's ticket history for the selected day or week, alongside its uptime and performance, so an outage and its ticket appear side by side, with time to resolution
Trust dashboards	Open and resolved tickets for the trust in the selected day or week, broken down by van



Needs attention 5 active

- Offline for 5 h 19 m** Derby / van2-blue
Modality not responding and PACS unreachable since 08:05. No GPS fix.
Ticket #4821, urgent, logged by Jo voice agent
- 3 failed image transfers** Bristol / sunflower
8 of 28 screened (29%). Running DEOS 32.0.
Ticket #4815, high
- Behind today's list** Liverpool / van3-bessie
4 of 26 screened (15%). 2 failed transfers.
Ticket #4819, high
- Behind today's list** Swindon / bali
1 of 8 screened (12%). Worklist missing on modality.
Ticket #4807, medium
- 5 transfers retrying** Reading / perky

Demo image: Ticket references alongside issues, including a ticket logged by Jo, the voice agent.

6.2 Opening a Ticket

- Click any ticket to open its detail in Deos Heartbeat: status, priority, requester, source, who it is assigned to, when it was created and last updated, and its activity timeline
- An Open in Freshdesk button takes you straight to that ticket in Freshdesk to reply, update, or close it
- A View van button takes you from the ticket to the van it relates to

Open tickets From Freshdesk

#4815, email High

Repeated sync failures

Retry queue partially cleared. Monitoring.

Assigned to Guna, updated yesterday 16:02

[Open in Freshdesk](#)

Demo image: A van's open tickets, with a direct link to Freshdesk.

- Tickets created by Jo, your voice agent, are matched to the correct trust and van automatically. Tickets from other channels that do not name a van appear as unassigned for you to link in one click

Tickets are view only in Deos Heartbeat. Replying to, updating, or creating tickets happens in Freshdesk, which keeps Freshdesk as the single system of record. Tickets refresh from Freshdesk every 15 minutes. Requires a Freshdesk API key from UKDEOS.



7. Email / Report Engagement Dashboard

Your reports are only valuable if they are read. The Email / Report Engagement Dashboard shows who is opening your daily and weekly reports, which trusts are engaged, and which recipients have gone quiet. It also brings your existing Wednesday Email Tracking report into Deos Heartbeat, so all reporting insight sits in one place.

7.1 What You Get

View	What You See
Overview	Reports sent this week, detected opens, and open rate compared with last week. An 8 week open rate trend, the least engaged trusts, and a list of recipients with no opens in four weeks or more
By Trust	For every trust: recipients, reports sent, detected opens, open rate, quiet recipients, and when a report was last opened
By Recipient	Everyone who receives reports, with reports received, opened, open rate, and last opened. Quiet recipients are clearly flagged
By Report	Every daily and weekly report run with its open rate. Select any report to see exactly who opened it and when
Wednesday Report	The existing Wednesday Email Tracking report: view the latest and past editions, turn it on or off, and manage its recipients

7.2 Acting on What You See

- Pause or remove a quiet recipient in one click, straight from the dashboard, using the same controls as Reporting Management
- Every change recorded in the audit trail
- Spot trusts where reports are not being read, and follow up before it becomes a problem at contract review

7.3 Engagement History

- Every report sent and every detected open is stored in Deos Heartbeat's own database, so engagement history builds up over time rather than being lost
- Weekly summaries keep every view fast, however much history has built up
- Combined with UKDEOS Owned Data, engagement can be viewed month on month and year on year, showing how actively trusts use their reporting

About detected opens: opens are detected when an email is opened and its content loads. Some email systems, which may include NHSmail, block this, which can undercount opens. Others open emails automatically, which can overcount them. Open figures are therefore presented as detected opens: a strong indicator of engagement rather than an exact count. The Wednesday report's existing content remains as it is today.



8. Trust Portal (Discovery Required)

A Trust Portal would give each trust's screening office secure access to live information about its own vans, turning the platform into a service UKDEOS can offer trusts directly. Because it involves external NHS organisations accessing a UKDEOS system, it needs to be designed around what trusts genuinely need and what their information governance teams will approve. For that reason it will be fully scoped following a discovery phase.

8.1 What the Portal Could Provide

- A live heatmap of the trust's own vans, with status, screening progress, and sync health
- Van detail for each of the trust's vans
- The trust dashboard, daily and weekly, with PDF export
- The fleet map, showing only that trust's vans

The final feature set will be confirmed by discovery. Some trusts may need more than this, and some may need less.

8.2 Why Discovery Comes First

- Trusts are separate NHS organisations, each with its own information governance, IT, and security requirements
- Different roles within a trust (screening office managers, superintendent radiographers, programme managers) are likely to need different things
- Compliance obligations for a system accessed by NHS staff need to be confirmed before design, not after
- How UKDEOS packages the portal commercially for trusts will shape what is built

8.3 Discovery: Understanding What Each Trust Needs

Structured conversations with a small number of trusts, selected with you, to answer:

- Who in the trust would use the portal, and what decisions would it help them make?
- Which information matters most: live van status, screening progress, weekly performance, location, or tickets?
- How often would they use it, and on which devices?
- Can the portal be reached from trust networks and devices, which often restrict external websites?
- Is the current PDF reporting sufficient for some trusts, and what would make live access worth having?
- Would trusts want to be notified directly when one of their vans has a problem?
- How should users be added and removed as staff join and leave?



8.4 Discovery: Compliance & Information Governance

A review of the governance requirements for giving trust staff access, carried out with UKDEOS and at least one trust information governance lead. Areas to assess:

Area	What Needs Confirming
Data classification	Confirm that the portal handles no patient identifiable data (only aggregate counts, device status, and staff user accounts) and document this for trusts
Data protection	Whether a Data Protection Impact Assessment is required, the data processing arrangements between UKDEOS, Thrumble, and each trust, and data retention
Data Security and Protection Toolkit	UKDEOS's current DSPT status, and whether trusts require it for this type of access
Supplier assurance	Whether trusts will require Cyber Essentials or Cyber Essentials Plus, independent penetration testing, or the NHS Digital Technology Assessment Criteria (DTAC)
Clinical safety	Confirm whether the portal falls outside the clinical safety standards (DCB0129 and DCB0160) as an operational rather than clinical system, and record the rationale
Identity and access	Whether trusts require NHS email based sign in, mandatory multi factor authentication, access approval processes, and access audit trails
Hosting and residency	UK data residency, availability expectations, backup, and incident response commitments
Accessibility	Accessibility standards expected by trusts, such as WCAG 2.2 AA

8.5 Discovery: Commercial & Operational Model

- How UKDEOS offers the portal to trusts: included in existing contracts or as a premium service
- Support model: who trust users contact, support hours, and onboarding
- Service levels UKDEOS would commit to for portal availability

8.6 What Discovery Delivers

- A requirements specification based on what trusts actually need
- A compliance plan setting out what is required, what is already in place, and what must be completed before launch
- A recommended feature set, security design, and support model
- A fully scoped proposal for the Trust Portal, including a pilot with one trust

Discovery is best carried out after the core platform is live, so trusts can see the real product during conversations.



9. Timeline

Add On	Requires	Confirmed by End of Discovery (Week 2)	Requested Later
Live Satellite Fleet Map	None	Within the 10 week programme	Around 1 week after go live
Location History (upgrade)	Live Satellite Fleet Map	Within 10 weeks, or up to 1 extra week	Around 2 weeks
Daily & Weekly Dashboards	None	Within 10 weeks, or up to 1 extra week	Around 2 weeks
UKDEOS Owned Data	Daily & Weekly Dashboards	Within the dashboards timeline	Around 1 week
Freshdesk Tickets: Open Tickets & History	Daily & Weekly Dashboards	Within the dashboards timeline	Around 1 week
Email / Report Engagement Dashboard	None	Within 10 weeks, or up to 1 extra week	Around 2 weeks
Trust Portal	Discovery first	After core platform go live	After core platform go live

How timelines work: add ons confirmed by the end of discovery are built alongside the core platform rather than after it. Most fit within the 10 week programme. Where a combination of add ons needs more time, the extension shown is the maximum, and the confirmed go live date is agreed at the end of discovery. Add ons requested after discovery are scheduled as a separate delivery after go live.

9.1 Support

- Add ons are covered by the same 1 month of post launch support for bug fixes and small amendments



10. Assumptions & Dependencies

Item	Assumption / Dependency
Timing of selection	Add ons confirmed by the end of discovery (Week 2) are built alongside the core platform. Add ons requested later are scheduled after go live
GPS data	Van locations are read from the GPS data already available in Grafana. Location accuracy varies with each van's surroundings and GPS signal, which is outside Thrumble's control. Vans without a working GPS fix will be flagged but cannot be plotted
History	Daily & Weekly Dashboards, UKDEOS Owned Data, and Location History import up to 120 days of existing history from Grafana at launch, subject to Grafana holding it. From then on, history is stored in Deos Heartbeat's own database
Map imagery	Satellite imagery is provided by a commercial map provider. Imagery detail varies by location and is outside Thrumble's control
Design	Add on screens are included in the two UI design sessions for the core platform
Grafana and third party software	As set out in the Deos Heartbeat Lite Scope of Work, Thrumble is not responsible for outages, data issues, or changes in Grafana or other third party software. Required updates are quoted as a change request
Email / Report Engagement	Depends on how opens are currently detected by the reporting and Wednesday automations, confirmed during discovery. Open figures are detected opens, as described in Section 7. The existing Wednesday report remains in place and continues to produce the same report
Freshdesk	UKDEOS provides a Freshdesk API key. Tickets are view only in Deos Heartbeat. Automatic van matching relies on the trust and van recorded on each ticket, as Jo already does. Replying to or updating tickets from Deos Heartbeat is not included
Trust Portal	Requirements and compliance will be defined through discovery with UKDEOS and selected trusts
Not included	Alerting and AI capabilities, which are described separately in the Deos Heartbeat Intelligence Roadmap

— END OF ADD ONS OVERVIEW —

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